

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The log obtained with the protocol network analyzer tcpdump is formed by UDP requests and ICMP responses, reflected in the 35084 substring of the log request and ICMP at the response. In particular, it is a DNS request (UDP 53), reflected at the response message "udp port 53 unreachable".

The log file indicates three attempts to resolve the DNS request, separated by line breaks. All of the requests are directed to the same ip address (203.0.113.2) from the same source (192.51.100.15:52444), all of them receiving the same ICMP answer "udp port 53 unreachable".

The foregoing is likely caused by a problem with the DNS server or the firewall related to it, which causes an inability of the server to receive, process and send the DNS response to the client.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred at 1:24 pm, when the cybersecurity analyst attempted to access the website www.yummyrecipesforme.com, receiving the error "destination port unreachable". The issue was first noticed when several customers of clients reported that they were not able to access the client company website www.yummyrecipesforme.com, and saw the error "destination port unreachable" after waiting for the page to load.

The IT team is currently investigating the issue and working to fix it. In our investigation, we discovered that the UDP port 53, which is used for DNS requests, is unreachable. The foregoing likely means the DNS server is down or has a huge load of traffic which makes it unable to process all the DNS requests or the firewall is blocking the requests to port 53. We must check the status of the DNS server and review the configuration of the firewall.

We are investigating the root of the problem, but likely the server was shutting down, suffered a DNS flood attack or there is a misconfiguration at the firewall.